

Julian Silva

*An Electronic and Telecommunication Engineering Undergraduate
at University of Moratuwa, Sri Lanka*

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OBJECTIVE

I am a dedicated student of Electronic and Telecommunication Engineering, driven by a profound passion for emerging technologies. My primary interests lie in machine learning, computer vision, embedded systems and software development. My academic journey has instilled in me a deep desire to contribute to innovative ideas. Committed to continuous self-improvement, I actively seek opportunities to enrich my skills. I am eager to collaborate with fellow experts to collectively advance the boundaries of knowledge and technology.

EDUCATION

- **University of Moratuwa, Sri Lanka** 2022 - Present
B.Sc. Honours in Electronic and Telecommunication Engineering (CGPA: 3.81/4.00)
Dean's List appearance in 5 out of 6 Semesters
- **St. Anthony's College, Kandy, Sri Lanka** 2007 - 2020
Passed GCE A/L Examination with 3A's in Physical Stream (*University Entrance Examination*) (Z-score: 2.3016)

WORK EXPERIENCES

- **Embedded & ML Engineering Intern**
Paraqum Technologies (Pvt) Ltd , Sri Lanka
 - **Tea Classification and Signature Estimation Using Computer Vision**
A computer-vision-based system that classifies tea grades and estimates visual signatures to support automated quality assessment.
Technologies: Python, TensorFlow, PyTorch
 - **PCB TestJig Firmware and Web Application**
Built the firmware and a Web Application for the PCB inspection device for Keyfobs by carrying out different types of tests.
Technologies: STM32 V3, STMCubeIDE, STMCubeProgrammer, Altium, Express.js, React.js, PostgreSQL
- **Robotics Instructor**
RoboticGen Academy (Pvt) Ltd, Sri Lanka
- **Physics Tutor for Edexcel Ordinary Level (IGCSE)**
Keenbridge English Academy (Pvt) Ltd.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, Kotlin, JavaScript
- **Machine Learning Libraries/Tools:** PyTorch, TensorFlow, Mediapipe OpenCV, YOLO, Scikit-learn
- **Embedded Systems Development Platforms :** ESP32, STM32, Raspberry Pi, Arduino
- **Electronic Design Automation Tools:** Matlab, Altium, LTspice, Verilog
- **Software Development Frameworks:** Node.js, React.js, React Native, Spring Boot
- **Computer-Aided Design Software and Robotics:** SolidWorks, ROS

PROJECTS

AI and Machine Learning

- **Neurosteered Target Speaker Extraction** Ongoing
Combines EEG signals with audio processing to isolate a listener's attended speaker in multi-speaker environments.
Technologies: Python, PyTorch, MATLAB, EEGLAB
- **Computer Vision-Based PCB Defects Detection Device** April 2024
A computer-vision system for detecting six PCB defect types and component-placement errors, integrated with a web interface. (*Individual Project*)
Technologies: Python, YOLOv8, PyTorch, OpenCV, Altium, SolidWorks
- **Panorama Stitching with Feature Matching Using Graph Neural Networks** November 2024
A panorama stitching system using SuperGlue for accurate feature matching and seamless image blending across changing viewpoints and lighting.
Technologies: SuperGlue, Python, OpenCV, PyTorch, RANSAC
- **Monocular RGB SLAM in Dynamic Environments** Ongoing
A robust monocular RGB SLAM system that uses uncertainty-aware tracking and mapping for dynamic object removal.
Technologies: PyTorch, Gaussian Splatting

Robotics

- **EKF SLAM in ROS** 2025
Implemented EKF-based SLAM in ROS for real-time localization and mapping with noisy sensor streams.
Technologies: ROS, Python, EKF
- **OboMouse - Micromouse Robot** (*MicroMaze 2024 – Champions*) 2024
An autonomous maze-solving robot using high-speed sensors and optimisation algorithms.
Technologies: STM32, C, Altium, SolidWorks
- **Robot Nanonaut** November 2023
A multifunction robot capable of line-following, obstacle avoidance, gripping, and sound recognition.
Technologies: Arduino, SolidWorks, Altium
- **SpeedObo - High-Precision Line Follower** Ongoing
A high-speed, high-accuracy line follower robot for competitive robotics.
Technologies: STM32, C++, Altium, SolidWorks
- **MetroniX Battlebot** (*UWV Robot Battle 2024*) February 2024
A combat robot featuring a dynamic lifting mechanism and RF-based control.
Technologies: Arduino, Altium, SolidWorks

Embedded Systems

- **Metroband - Vibrating Tempo-Keeping Wristband** (*Mora Ventures 6.0*) August 2023
A wearable tempo-keeping wristband for musicians with vibration cues and beat detection.
Technologies: C++, Altium, SolidWorks, Python
- **Analog Guitar Pedal Board** December 2023
A fully analogue guitar effects pedal board featuring six sound-processing modules.
Technologies: Analog Circuits, LTspice, Altium

Software Development

- **Sportsense - AI Sports Trainer** (*Idealize 2024 – 1st Runner-Up*) July 2024
A real-time CV-based posture and movement analyser offering personalised sports training feedback.
Technologies: Kotlin, Firebase, Mediapipe, YOLO
- **Signify - Speech-to-Sign-Language AR Glasses** (*VR Cade 2025 – Champions*) January 2025
An AR-based system translating speech into sign language in real-time for hearing-impaired users.
Technologies: Unity, Blender, OpenCV, Mediapipe
- **PMF Analysis Agent** 2025
An AI-driven Product-Market Fit analyser for SaaS companies, evaluating user traffic, reviews, and behavioural patterns to generate actionable insights.
Technologies: React, Node.js, LangGraph, Python, AWS
- **Xplore - Travel Companion App** 2024
A mobile application providing personalised travel planning and location-based insights.
Technologies: Next.js, React, Firestore, Firebase Storage

- **Data Analysis Agent** 2025
An intelligent agent for automated analysis and reporting on financial data.
Technologies: React, AWS, Express.js, Python, LangGraph
- **Crystal Clear - Platform for Assisting Dyslexic Students** (*Imagine Cup 2024*) January 2024
An AI-powered personalised learning platform using computer vision and voice recognition for dyslexic students.
Technologies: React, React Native, TensorFlow
- **PCB Company Web Application** May 2024
A full-stack web system with six subsystems for managing PCB company operations.
Technologies: Java, Spring Boot, MongoDB, React

SOFT SKILLS

- Leadership Skills
- Logical Thinking & Analytical Skills
- Communication & Interpersonal Skills
- Presentation Skills
- Language Skills: English, Sinhalese

ACHIEVEMENTS

Technical & Innovation Competitions

- Champions - VR Cade 2025 (Mixed Reality Competition), IEEE IAS at UoM
- Champions - MicroMaze 2024 (Micromouse), IEEE RAS at IIT
- 1st Runner-Up - Idealize 2024 (Ideathon & Mobile App), AIESEC UoM
- Conceptualist Award - Idealize 2024 (AIESEC UoM)
- 2nd Runner-Up - Robofest 2024 (Micromouse), SLIIT
- 5th Place - SPARK 2024 Innovation Competition, Electronic Club UoM
- 7th Place - UWU Robot Battle Competition
- Finalist - Brainstorm 2024 (Sri Lanka's Premier Healthcare Innovation Competition)
- Finalist - IESL Startup Challenge 20205
- Semifinalist - Microsoft ImagineCup 2024 (AI-Driven Student Startups)

Academic Excellence

- Dean's list placement (Semesters 1, 2, 4, 5 and 6), University of Moratuwa
- Winner - Best Student in Physics of the Year (2020), St. Anthony's College Kandy
- Second Place - Inter-School Mathematics Competition (Mahamaya College Kandy)

Leadership

- Best Young Leader Nominee - AIESEC Sri Lanka (2022)
- Spirit of Incoming Global Volunteering Nominee - AIESEC Colombo South (2022)
- Best Team Leader & Best Project of the Quarter - AIESEC UoM (IGV)
- Second Best Leader - AMPLIFIER 2021, AIESEC Sri Lanka
- Spirit of the Month - AIESEC Colombo South (March 2023)
- Most Impactful Project of the Term - IDEALIZE 2022 (AIESEC Colombo South)

Music & Performing Arts

- Finalist - Chaturmana '23 (Inter-University Talent Competition) – Choir TripleG
- Winner - Best Musician of the Year (2019), St. Anthony's College Kandy
- First Place - All-Island Choir Competitions (2019, 2017, 2015)
- Second Place - All-Island Choir Competitions (2018, 2016)

Sports

- Runner-Up - Inter-School Basketball Tournament (2016)
- Runner-Up - Zonal Chess Championship (2015)

EXTRA-CURRICULAR

University Societies & Technical Leadership

- Head of External Relations - Electronic Club, University of Moratuwa
- Executive Committee Member - IEEE Industrial Electronics Society, University of Moratuwa
- Industrial Relations Manager - Electronic Club, University of Moratuwa
- Committee Member - Asia-Oceania Top University League on Engineering (AOTULE) Conference 2023
- Programme Committee Co-Lead - MoraForesight 1.0
- Logistics Team Leader - AQTULE International Deans' Conference 2023 (Faculty of Engineering)
- Active Member - Gavel Club, Classical Music Society, Mora Hiking Club

AIIESEC

- Organizing Committee Vice President - National Leadership Seminar 2022 (AIIESEC Sri Lanka)
- AIIESEC International Congress Committee Member (Asia Pacific Summit 2022)
- Organizing Committee Member - IDEALIZE 2022 (AIIESEC Colombo South)

Public Speaking & Event Hosting

- Hosted major UoM events including Colours Awards 2023/2024, CS Eminence 2023, MoraForesight 1.0 Closing Ceremony, SLUG Baseball Ceremonies, and Global Village 2023
- Host - BSc Cosmetology Graduation Ceremony 2022 (Academy of Universal Global Peace USA)

Music & Cultural Leadership

- Choir Master - Catholic Students' Movement, University of Moratuwa (2023)
- Choir Master- St. Anthony's College, Kandy (2018/2019)

School Leadership & Sports

- College Senior Prefect - St. Anthony's College (2018/2019)
- Vice President of the English Literary Circle - St. Anthony's College (2018/2019)
- Assistant Treasurer of the Media Club - St. Anthony's College (2018/2019)
- Captain - College Chess Team (2015)
- College Basketball Team Player

REFERENCES

Dr. Ranga Rodrigo

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